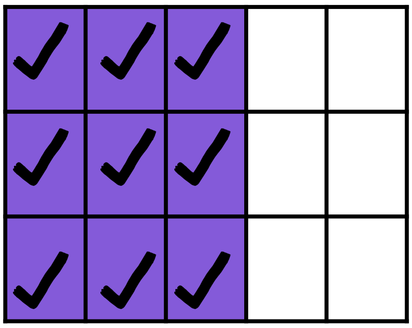
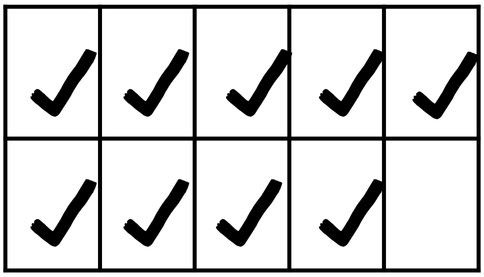
Dividing a fraction by a fraction

Task A: Dividing two fractions

1) Work out the answer to the following. You can use these templates for the area model if it helps.

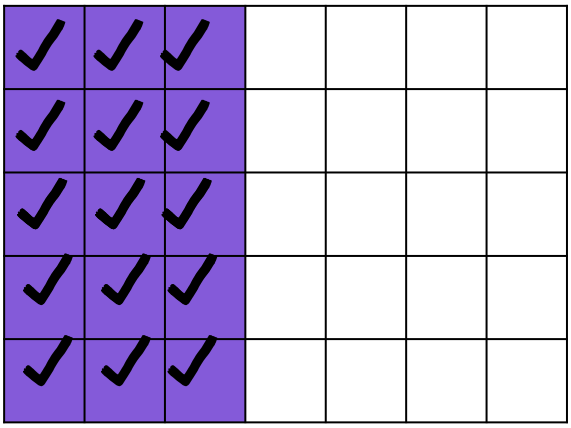
$$\frac{3}{5} \div \frac{2}{3} = \frac{9}{10}$$

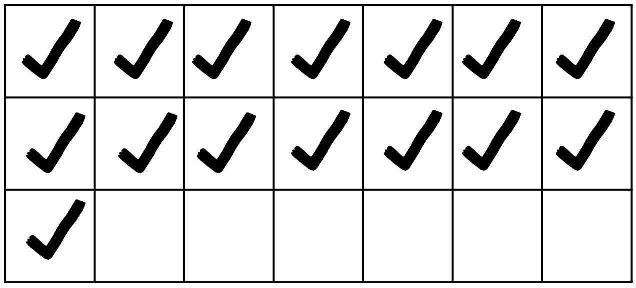




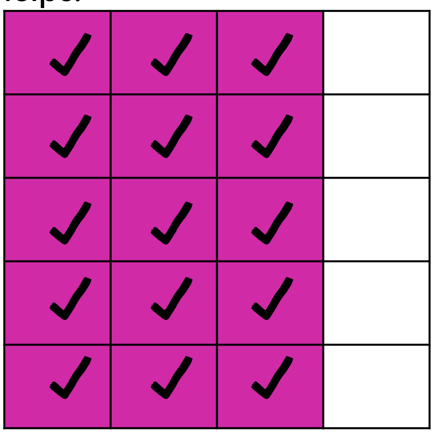
2) Work out the answer to the following. You can use these templates for the area model if it helps.

$$\frac{3}{7} \div \frac{3}{5} = \frac{15}{21} = \frac{5}{7}$$

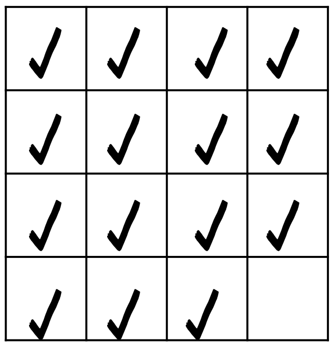




3) Work out the answer to the following by drawing area models. Use squared paper if it helps.



$$\frac{3}{4} \div \frac{4}{5} = \frac{15}{16}$$



Task B: Reciprocals

1) Work out the **reciprocals** of the following.

a) 8 **reciprocal** is $\frac{1}{8}$

b) 10 **reciprocal** is $\frac{1}{10}$

c) 7 **reciprocal** is $\frac{1}{7}$

d) -3 **reciprocal** is $-\frac{1}{3}$

e) $\frac{11}{3}$ **reciprocal** is $\frac{3}{11}$

f) $\frac{2}{9}$ **reciprocal** is $\frac{9}{2}$

g) $\frac{4}{15}$ **reciprocal** is $\frac{15}{4}$

h) 1 **reciprocal** is 1

2) Work out the reciprocals of the following.

a) $5\frac{1}{2}$ $\frac{11}{2}$ **reciprocal** is $\frac{2}{11}$

b) $9\frac{2}{3}$ $\frac{29}{3}$ **reciprocal** is $\frac{3}{29}$

c) $10\frac{3}{5}$ $\frac{53}{5}$ **reciprocal** is $\frac{5}{53}$

3) Using the cards only once, make the following statements true.

| 1 | 2 | 2 | 4 | 5 | 8 | 10 |

E.g.

The **reciprocal** of $\frac{1}{2}$ is 2

The **reciprocal** of $\frac{4}{5}$ is $\frac{10}{8}$

This is an equivalent fraction to the **reciprocal**

Task C: Division by multiplying by the reciprocal

1) Work out the answers to the following, simplifying your answer where necessary.

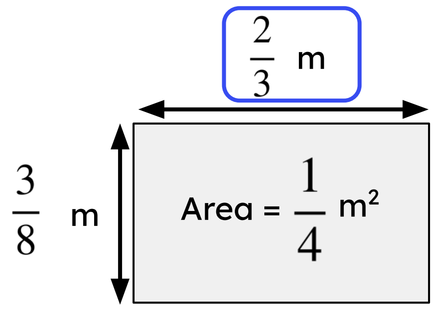
a) $\frac{3}{7} \div \frac{4}{9} = \frac{3}{7} \times \frac{9}{4} = \frac{27}{28}$

b) $\frac{9}{10} \div \frac{8}{11} = \frac{9}{10} \times \frac{11}{8} = \frac{99}{80} = 1\frac{19}{80}$

c) $\frac{3}{4} \div \frac{5}{12} = \frac{3}{4} \times \frac{12}{5} = \frac{36}{20} = \frac{9}{5} = 1\frac{4}{5}$

2) Here are some areas of some shapes. Work out the missing lengths

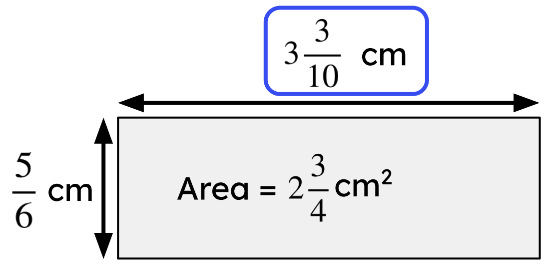
a)



$\frac{1}{4} \div \frac{3}{8} = \frac{1}{4} \times \frac{8}{3}$

$= \frac{8}{12} = \frac{2}{3}$

b)

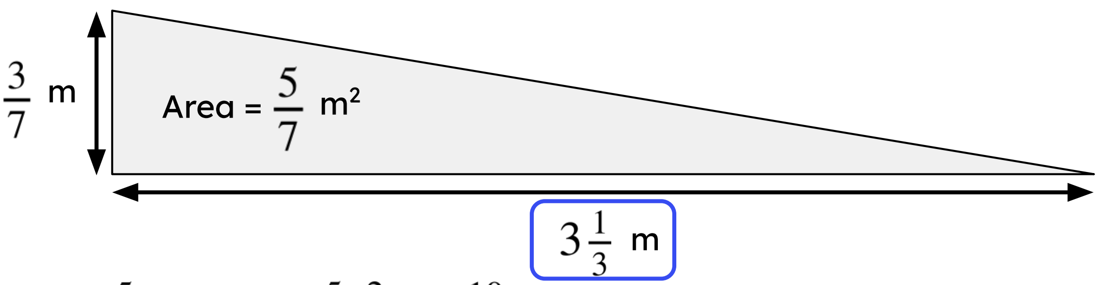


$2\frac{3}{4} \div \frac{5}{6} = \frac{11}{4} \times \frac{6}{5}$

$= \frac{66}{20} = 3\frac{6}{20}$

$= 3\frac{3}{10}$

c)



$\frac{5}{7} \times 2 = \frac{5 \times 2}{7} = \frac{10}{7}$

$\frac{10}{7} \div \frac{3}{7} = \frac{10}{7} \times \frac{7}{3}$

$= \frac{70}{21} = \frac{10}{3} = 3\frac{1}{3}$